Week 11

Spring 2027 · Section 029 · MW 9:00–10:20 AM · RRC Room 8210

Course Level Objectives:

- Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. (CLO 3)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe how to generate randomization samples for a given study design. (CLO 8) **2.** Identify the centers of the bootstrap and randomization distributions for given data. (CLO 5) **3.** Use a 95% confidence interval to reach a two-tailed test conclusion at α = 0.05. (CLO 6, CLO 8) **4.** Determine whether a confidence interval or a hypothesis test fits a research question. (CLO 6, CLO 8)	e-Text: 4.5 Making Connections Interactive Lecture: 4.5 Making Connections Slides: 4.5 Making Connections	Participation: Making Connections – body temperature: test 98.6 using the CI (98.05, 98.47); sleep study (28/50): how to generate randomization samples Your Turn: CI or Test? (smart-speaker ownership)	Online Homework: 4.5 Making Connections (due Sun Apr 11, 11:59 PM) Take-Home Quiz 7 (due Mon Apr 12, 11:59 PM) – covers 4.4, 4.5, 5.1 (all objectives) Test 3 (Fri Apr 30 – Sun May 2, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Apr 29) (all objectives)	WileyPLUS StatKey Blackboard Respondus
5. Explain when a normal distribution can replace a randomization distribution. (CLO 5) **6.** Compute a standardized z statistic. (CLO 8) **7.** Find a p-value as a tail area under the standard normal curve. (CLO 3, CLO 8)	e-Text: 5.1 Hypothesis Tests Using Normal Distributions Interactive Lecture: 5.1 Hypothesis Tests Using Normal Distributions Slides: 5.1 Hypothesis Tests Using Normal Distributions StatKey – Theoretical Distributions (Normal)	Participation: Normal Distributions and z – malaria/mosquito example: compute z, shade the tail in StatKey, find the p-value Your Turn: z-Tests (right-, left-, and two-tailed)	Online Homework: 5.1 Hypothesis Tests Using Normal Distributions (due Sun Apr 11, 11:59 PM) Take-Home Quiz 7 (due Mon Apr 12, 11:59 PM) – covers 4.4, 4.5, 5.1 (all objectives) Test 3 (Fri Apr 30 – Sun May 2, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Apr 29) (all objectives)	WileyPLUS StatKey Blackboard Respondus

Also this week: Test 2 Retake – Mon Apr 5 – Sun Apr 11 (optional; higher score counts)